

Assignment 7

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Title

Perform various Data Transformation and Error Correction techniques on the datasets.

Aim

Transform raw Titanic via error correction (dedupe/impute), outlier handling (IQR cap 116 Fare), feature engineering (log skew reduction 4.79→0.39, Age bins), scaling (MinMax/Z-score), encoding (Sex labels).

Theory

IQR outliers: Clip beyond $Q1 - 1.5IQR$ to $Q3 + 1.5IQR$; reduces extremes.

$\log(1+x)$: Skew → symmetric for positives; Z-score standardizes $\mu=0$ $\sigma=1$.

Binning discretizes continuous; LabelEncoder maps categories ordinally.

Real-world datasets frequently contain **errors, missing values, outliers, and inconsistent formats**. Data transformation and error correction are crucial for improving data quality and ensuring that statistical procedures and ML models behave reliably.

Error correction includes detecting duplicated rows, inconsistent category labels, and impossible values (e.g., negative Age). Duplicates are removed to avoid over-representing repeated records. Missing values can arise from measurement issues or recording errors and are commonly handled via imputation. Group-wise imputation, such as filling Age with the median within each Pclass, preserves **structure and relationships** in the data better than a global mean.

Outlier treatment uses methods like IQR clipping to reduce the influence of extreme values. For heavily skewed variables like Fare, **log transformation** ($\log(1+x)$) stabilizes variance and makes distributions more symmetric, which improves the performance of models that assume roughly normal residuals. **Scaling** (MinMax or Z-score) brings features to comparable ranges, preventing variables with large magnitudes from dominating others in distance-based or gradient-based algorithms. **Encoding** categorical variables (e.g., LabelEncoder for Sex, one-hot encoding for Embarked) converts text labels into numeric form while preserving meaning. Together, these steps produce a **clean, well-behaved feature matrix** suitable for downstream analysis and modelling.

Dataset

Raw Titanic (891×12): Age 19.87% missing, Cabin 77.10%, Embarked 0.22%; Fare skew=4.79, 116 outliers.

Technique	Target	Effect
Impute	Age/Emb/Fare	0 missing numerics
Cap	Fare	116 outliers bounded
Log	Fare	Skew 4.79 → 0.39
Scale	Age/Fare	/ Z(0,1)
Encode/Bin	Sex/Age	Numeric ready

Code

Data Science Lab Assignment 7: Data Transformation & Error Correction

Dataset: Raw Titanic

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler, MinMaxScaler, LabelEncoder
from scipy import stats
import warnings; warnings.filterwarnings('ignore')

df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

print("Original shape:", df.shape)
print("Missing %%:\n", (df.isnull().mean() * 100).round(2))

# 1. ERROR CORRECTION: Duplicates
print("\nDuplicates:", df.duplicated().sum())
df_clean = df.drop_duplicates().reset_index(drop=True)
```

2. MISSING IMPUTATION

```
df_clean['Age'] = df_clean.groupby('Pclass')['Age'].transform(lambda x: x.fillna(x.median()))  
df_clean['Embarked'] = df_clean['Embarked'].fillna(df_clean['Embarked'].mode()[0])  
df_clean['Fare'] = df_clean['Fare'].fillna(df_clean['Fare'].median())  
print("Missing after:", df_clean.isnull().sum().sum())
```

3. OUTLIER CAP IQR (Fare)

```
Q1, Q3 = df_clean['Fare'].quantile([0.25, 0.75])  
IQR = Q3 - Q1  
bounds = [Q1 - 1.5 * IQR, Q3 + 1.5 * IQR]  
outliers = ((df_clean['Fare'] < bounds[0]) | (df_clean['Fare'] > bounds[1])).sum()  
df_clean['Fare_cap'] = df_clean['Fare'].clip(bounds[0], bounds[1])  
print("Fare outliers:", outliers)
```

4. TRANSFORMATIONS

```
df_clean['Fare_log'] = np.log1p(df_clean['Fare'])  
df_clean['Age_bin'] = pd.cut(df_clean['Age'], bins=5, labels=['Ch', 'Te', 'Yo', 'Ad', 'Sr'])
```

5. SCALING

```
mm = MinMaxScaler(); df_clean['Age_mm'] = mm.fit_transform(df_clean[['Age']]).flatten()  
z = StandardScaler(); df_clean['Fare_z'] = z.fit_transform(df_clean[['Fare_cap']]).flatten()
```

6. ENCODING

```
le = LabelEncoder(); df_clean['Sex_le'] = le.fit_transform(df_clean['Sex'])  
  
print("\nSkew Fare: %.2f -> log: %.2f" % (df_clean['Fare'].skew(), df_clean['Fare_log'].skew()))  
print("Age range: %.0f -> MM[0,1]" % (df_clean['Age'].max() - df_clean['Age'].min()))  
print("Fare_z:  $\mu$ =%.3f  $\sigma$ =%.3f" % (df_clean['Fare_z'].mean(), df_clean['Fare_z'].std()))  
print("\nTransformed sample:")  
print(df_clean[['Age', 'Age_mm', 'Fare', 'Fare_log', 'Sex', 'Sex_le', 'Age_bin']].head())
```

```
print("\nSummary:\n- Dupes removed\n- Impute Age(grp med), Emb(Mode), Fare(med)\n- Cap %d Fare outliers\n- Log Fare, bin Age\n- Scale MM(Age)/Z(Fare)\n- Label Sex"%outliers)
```

Terminal Output

Original shape: (891, 12)

Missing %:

PassengerId 0.00

Survived 0.00

...

Age 19.87

...

Cabin 77.10

Embarked 0.22

Duplicates: 0

Missing after: 687

Fare outliers: 116

Skew Fare: 4.79 -> log: 0.39

Age range: 80 -> MM[0,1]

Fare_z: $\mu=0.000$ $\sigma=1.001$

Transformed sample:

	Age	Age_mm	Fare	Fare_log	Sex	Sex_le	Age_bin
0	22.0	0.271174	7.2500	2.110213	male	1	Te
1	38.0	0.472229	71.2833	4.280593	female	0	Yo
2	26.0	0.321438	7.9250	2.188856	female	0	Te
3	35.0	0.434531	53.1000	3.990834	female	0	Yo
4	35.0	0.434531	8.0500	2.202765	male	1	Yo

Summary:

- Dupes removed
- Impute Age(grp med), Emb(Mode), Fare(med)
- Cap 116 Fare outliers
- Log Fare, bin Age
- Scale MM(Age)/Z(Fare)
- Label Sex

Observations

No duplicates; imputation handles all but Cabin (ignored); 116/891=13% Fare outliers capped.

Log transforms extreme skew; scales normalize ranges; bins/encoding enable modeling.

Conclusion

Full pipeline yields production-ready Titanic: error-free, transformed, scaled.